# Essential Summary

**Padé**

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*, Vol. 45, No. 3, 461–464 (2013) DOI: 10.6052/0459-1879-12-277*



## 1 Background and Motivation

The determination of homoclinic and heteroclinic orbits is central to understanding global bifurcations, chaos, and solitary wave solutions in nonlinear dynamics. Classical methods—such as Melnikov's method, hyperbolic perturbation, and Padeé approximation—have been applied with varying success. However, existing Padeé-based approaches share a common limitation: the numerator and denominator of the approximant are restricted to polynomial functions, which limits their flexibility in capturing the exponential decay behavior characteristic of homoclinic orbits.

This paper introduces the **generalized Padé approximation (GPA) method**, a theoretical extension that removes this restriction by allowing the approximant to be constructed from *any* kind of function series. This provides a unified framework that encompasses all existing Padeé-type modifications as special cases.

## 2 Core Innovation: The Generalized Padé Approximation

### 2.1 From Classical to Generalized

The classical Padé approximant of a formal power series $f(z) = \sum_{i=0}^{\infty} a_i z^i$ is a rational function $P_L(z)/Q_M(z)$ where $P_L \in H_L, Q_M \in H_M$ (polynomials of degree $L, M$), satisfying:

$$f(z) - \frac{P_L(z)}{Q_M(z)} = \mathcal{O}(z^{L+M+1}), \qquad Q_M(0) = 1. \tag{1}$$

The generalized definition replaces the polynomial spaces $H_L$ and $H_M$ with extended function spaces:

$$\widehat{H}_m = \left\{P : P(z) = \sum_{i=0}^{m} a_i g_i(z)^i,\ g_i : \mathbb{C} \to \mathbb{C},\ a_i \in \mathbb{C}\right\}, \tag{2}$$

$$G(m, n) = \left\{R = P/Q : P \in \widehat{H}_m,\ Q \in \widehat{H}_n \setminus \{0\}\right\}. \tag{3}$$

**Definition (Generalized Padeé Approximation):** If there exists a rational function $P_L/Q_M \in G(L, M)$ satisfying

$$f(z) - \frac{P_L(z)}{Q_M(z)} = \mathcal{O}(z^{L+M+1}), \tag{4}$$

then $P_L/Q_M$ is called the **Generalized Padeé Approximant**, denoted $\text{GPA}[L/M]_f$.

When $g_i(z) = z^i$ (the classical polynomial basis), the GPA reduces to the classical Padeé approximant.

### 2.2 Homoclinic Orbit Construction via Hyperbolic Functions

For the autonomous oscillator $\ddot{x} + g(x) = 0$, assume the power series solution:

$$x(t) = \sum_{i=0}^{\infty} a_i t^i, \tag{5}$$

where the coefficients $a_i$ are determined by substituting into the equation and matching powers. For a homoclinic orbit through the saddle $H(h, 0)$, the initial condition $a = x(0)$ is determined by the energy integral $\int_h^a g(x)dx = 0$, with $a_1 = \dot{x}(0) = 0$.

Since $\lim_{t\to\pm\infty} x(t) = h$, a power series representation fails to capture the asymptotic behavior. The key innovation: construct a GPA using **hyperbolic functions** that naturally satisfy the boundary conditions:

$$\boxed{\mathrm{GPA}[L/M] = \frac{\displaystyle\sum_{i=0}^{L} \alpha_i \operatorname{sech}^i t}{\displaystyle 1 + \sum_{i=1}^{M} \beta_i \operatorname{sech}^i t}}. \tag{6}$$

The $L + M + 1$ unknown coefficients $\{\alpha_i, \beta_i\}$ are determined by expanding the GPA into a Taylor series around $t = 0$ and matching the first $L+M+1$ terms with the power series coefficients $\{a_i\}$, together with the boundary condition at infinity (automatically satisfied by $\operatorname{sech}^i t \to 0$ as $t \to \pm\infty$).

### 2.3 Advantages of the Hyperbolic-Series Approximant

1. **Simpler Taylor expansion:** $\operatorname{sech}^i t$ has a simpler Maclaurin series than exponentials, reducing computational burden at the same approximation order.
2. **Automatic boundary condition satisfaction:** The $\operatorname{sech}^i t \to 0$ behavior at infinity eliminates the need to enforce boundary conditions separately.
3. **Generality:** The method is applicable to *any* $g(x)$ that is a polynomial or rational function—not restricted to specific oscillator families.

## 3 Key Results: Numerical Examples

### 3.1 Example 1: Helmholtz–Duffing Oscillator, $g(x) = c_1 x + c_2 x^2$

$$\ddot{x} + c_1 x + c_2 x^2 = 0. \tag{7}$$

With $c_1 = -3, c_2 = 100$ (strongly asymmetric, large nonlinearity), $L = M = 2$:

$$\mathrm{GPA}[2/2] = \frac{0.005 \operatorname{sech} t + 0.07 \operatorname{sech}^2 t}{1 + 0.611111 \operatorname{sech} t + 0.055555 \operatorname{sech}^2 t}. \tag{8}$$

The phase portrait matches Runge–Kutta with excellent agreement despite $c_2 = 100 \gg 1$.

### 3.2 Example 2: Full Helmholtz–Duffing, $g(x) = c_1 x + c_2 x^2 + c_3 x^3$

$$\ddot{x} + c_1 x + c_2 x^2 + c_3 x^3 = 0. \tag{9}$$

With $c_1 = -2, c_2 = 5, c_3 = 15$, both left and right homoclinic orbits are captured simultaneously by solving for $\{\alpha_i, \beta_i\}$ with $L = M = 2$, demonstrating that the GPA handles asymmetric cubic–quadratic nonlinearity without modification.

### 3.3 Example 3: Duffing–Harmonic Oscillator, $g(x) = \frac{\lambda x+\mu x^3}{1+\nu x^2}$

$$\ddot{x} + \frac{\lambda x + \mu x^3}{1 + \nu x^2} = 0. \tag{10}$$

With $\lambda = -1, \mu = 1, \nu = 1$ and $L = M = 3$, the GPA captures homoclinic orbits of this rational-restoring-force oscillator—a system whose exact solution cannot be expressed in terms of elementary or classical elliptic functions.

## 4 Significance

1. **Theoretical unification:** The GPA definition unifies all existing Padeé-type modifications (quasi-Padeé, modified Padeé, exponential-function approximants) under a single theoretical framework—previous works become special cases of the GPA.
2. **Computational efficiency:** The $\operatorname{sech}^i t$ basis produces simpler Taylor expansions than exponential-based alternatives, reducing the cost of determining the $\{\alpha_i, \beta_i\}$ coefficients at the same approximation order.
3. **Broad applicability:** The method handles polynomial $g(x)$ of arbitrary degree and rational $g(x)$ of the form $(\lambda x + \mu x^3)/(1 + \nu x^2)$ without modification—a significant advantage over methods restricted to specific restoring force forms.

## 5 Limitations

- The method solves for homoclinic orbits of conservative systems only; damped systems require separate treatment.
- The GPA approximation order $L = M$ must be chosen empirically; no a priori error estimate is provided.
- Only single-degree-of-freedom autonomous oscillators are considered.

## 6 Relationship to the MGHFP Series

This paper and its English counterpart (Chinese Physics B 2014) are methodologically distinct from the GHFP/MGHFP series—the GPA constructs homoclinic solutions directly via rational function approximation, whereas GHFP/MGHFP uses perturbation techniques to derive amplitude–parameter relationships. However, both share the common goal of **analytical treatment of strongly nonlinear oscillators**. The $\operatorname{sech}^i t$ basis introduced in this paper later inspired the construction of homoclinic solution forms ($x = a \sin^2 \varphi + b$) in the GHFP framework (JSV 2013).

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